

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection MATLAB 2018a, 2018b, 2019a, ScanImage 2018, Fiji (<https://fiji.sc>), FicTrac v2.0 and v2.1 (<https://github.com/rjdmoores/fictrac>), neuprint (<https://neuprint.janelia.org/>), Python3, R, NeuprintR 1.1 (<https://github.com/natverse/neuprintR>) and natverse (<https://github.com/natverse/natverse>)

Data analysis Motion correction was performed using NoRMCorre. Analysis of calcium imaging and electrophysiology data was performed using custom code written in MATLAB. Confocal images were analyzed using Fiji (Image J). Computational modeling and analyses were performed in Python.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	All sample sizes were chosen based on conventions in our field for standard sample sizes. These sample sizes are conventionally determined on the basis of the expected magnitude of animal-to-animal variability, given published results and pilot data.
Data exclusions	For calcium imaging data analysis for closed loop walking behavior without loom stimulus, no flies were excluded from the dataset. For calcium imaging data analysis for loom experiments, we excluded flies where fluorescence was too dim or when the bump position offsets from heading cue position during forward walking bouts were highly unstable (1/11 flies in EPG dataset, and 6/17 flies in the hΔB dataset). We identified large backward walking epochs in 5/10 remaining flies in the EPG dataset and 10/11 flies in the hΔB dataset. For electrophysiology analysis, we excluded experiments if the fly did not sample the full 360-degree heading range, if there was large electrical noise, or if the fly's total speed was not above a minimum threshold of 0.5 mm/s for over 20% the total experimental period. This occurred in 14/28 cells recorded; we included 14 cells across 11 flies in our dataset.
Replication	For all experiments, results were replicated in different individual flies across each dataset. We did not omit any replicates on the basis of the experimental result. A few flies (or trials) were excluded due to factors that prevented us from analyzing the data -- e.g., the fly simply did not walk; all these cases of data exclusion are noted explicitly in the Online Methods.
Randomization	For optogenetic activation experiments (Fig. 2f, Extended Data Fig. 7d) and for behavioral experiments (Fig. 3a-e) flies were grouped for analysis based on genotype. Flies were never arbitrarily assigned to treatment groups, and therefore there were no experiments where randomization could have been performed.
Blinding	The experimenter was not blind to genotype in this study. This is because the different genotypes in the study were used to target a genetically encoded calcium indicator to different cell types, and so the genotype of the flies was obvious during the course of the imaging experiment, based on the imaged pattern of fluorescence.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

chicken anti-GFP (Abcam), Cat# ab13970
 mouse anti-Bruchpilot (Developmental Studies Hybridoma Bank), Cat# nc82
 Alexa Fluor 488 goat anti-chicken (Invitrogen), Cat# A11039
 Alexa Fluor 633 goat anti-mouse (Invitrogen), Cat# A21050
 rat anti-FLAG (Novus Biologicals), Cat# NBP106712B
 rabbit anti-HA (Cell Signal Technologies), Cat# 3724S
 DyLight 550 mouse anti-V5 (AbD Serotec), Cat# MCA1360D550GA
 Alexa Fluor 488 goat anti-rabbit (Invitrogen), Cat# A11008
 ATTO 647 goat anti-rat (Rockland), Cat# 612-156-120
 Alexa Fluor 405 goat anti-mouse (Invitrogen), Cat# A175660

Validation

The anti-GFP antibody (Adcam) is the standard antibody used in the field for labeling Green Fluorescent Protein (GFP). The anti-bruchpilot antibody (nc82, DSHB) is a standard in the field as a background stain that labels presynaptic active zones. The secondary antibody we used to label GFP expressing cells (Alexa Fluor 488 goat anti-chicken) was verified by us to target only those cells which express live GFP fluorescence. The secondary antibody used for background (neuropil) staining (Alexa Fluor 488 goat anti-chicken, Alexa Fluor goat anti-mouse 633) was verified by us to reproduce the known patterns of neuropil borders (nc82 immunoreactivity) in published atlases (VirtualFlyBrain.org). Antibodies used for MCFO immunostaining (rat anti-FLAG, rabbit anti-HA, DyLight 550 mouse anti-V5, AlexaFluor 488 goat anti-rabbit, ATTO 647 goat anti-rat) are validated in *Drosophila melanogaster* for this application in Nern et al., 2015.

Animals and other organisms

Policy information about [studies involving animals](#): [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals

For calcium imaging experiments, we used female flies 20-50 hours post-eclosion. For optogenetic activation experiments, we used female flies 1-5 days post-eclosion, while for all other electrophysiology experiments, we used female flies 24-48 hours old. For behavior experiments, we used 3-5 day old female flies.

The following stocks were obtained from the Bloomington *Drosophila* Stock Center (BDSC) and published as follows: P{GMR60D05-GAL4}attP2 (BDSC 39247), P{GMR16D01-lexA}attP40 (BDSC 52503), P{R72B05-p65.AD}attP40 (BDSC 70939), P{VT055827-GAL4.DBD}attP2 (BDSC 71851), P{VT008681-GAL4.DBD}attP2 (BDSC 73701), Mi{Trojan-p65AD.2}Vglut[Mi04979-Tp65AD.2] (BDSC 82986), PBac{20XUAS-IVS-jGCaMP7f}VK00005 (BDSC 79031), and P{p65.AD.Uw}attP40; P{GAL4.DBD.Uw}attP2 (BDSC 79603). MCFO experiments used w[1118], P{R57C10-FLPL}su(Hw)attP8; +; PBac{10xUAS(FRT.stop)myr::smGdP-HA}VK00005, P{10xUAS(FRT.stop)myr::smGdP-V5-THS-10xUAS(FRT.stop)myr::smGdP-FLAG}su(Hw)attP1 (BDSC 64087) and w[1118], P{R57C10-FLPG5}su(Hw)attP8; +; PBac{10xUAS(FRT.stop)myr::smGdP-HA}VK00005, P{10xUAS(FRT.stop)myr::smGdP-V5-THS-10xUAS(FRT.stop)myr::smGdP-FLAG}su(Hw)attP1 (BDSC 64088).

The split-Gal4 line targeting PFNd neurons was ss00078 (P{R16D01-p65.AD}attP40; P{R15E01-GAL4.DBD}attP2). The split-Gal4 line targeting SpsP neurons was ss52267 (P{VT019012-p65.AD}attP40; P{R72C10-GAL4.DBD}attP2). The split-Gal4 line targeting IbSpsP neurons was ss04778 (P{R47G08-p65.AD}attP40; P{VT012791-GAL4.DBD}attP2). The split-Gal4 line targeting PFNV neurons was ss52628 (P{R22G07-p65.AD}attP40; P{VT063307-GAL4.DBD}attP2). The split-Gal4 line targeting LNO1 neurons was ss47398 (P{VT020742-p65.AD}attP40; P{VT017270-GAL4.DBD}attP2). These lines were obtained from the Janelia Research Campus FlyBank and have been described previously.

P{20XUAS-IVS-mCD8::GFP}attP40 was a gift from Barret Pfeiffer and Gerald Rubin and was described previously. The recombinant chromosome P{13xLexAop2-IVS-pmyr::GFP}VK00005, P{20xUAS-CsChrimson-mCherry-trafficked}su(Hw)attP1 was a gift from Vivek Jayaraman. Gr43a-LexA was a gift from Hubert Amrein and was described previously. 13XLexAop2-IVS-Syn21-Chrimson::tdT-3.1-p10-F8 (VK00005) was a gift from Barret Pfeiffer and David Anderson and was described previously. P{10xUAS-IVS-hKCNJ2.EGFP}attP2 was a gift from Gwyneth Card (via Barret Pfeiffer and Gerry Rubin) and was described previously.

We constructed a split-Gal4 line to target LNO2 neurons that incorporates the VglutAD transgene. This split-Gal4 line is +;Mi{Trojan-p65AD.2}Vglut[Mi04979-Tp65AD.2]; P{VT008681-GAL4.DBD}attP2. We validated the expression of this line using immunohistochemical anti-GFP staining, and also using Multi-Color-Flip-Out (MCFO) to visualize single-cell morphologies. On occasion, this split line labels a cell type innervating nodulus subunit 3 (NO3); MCFO results suggest that this is a separate cell type from LNO2 and does not innervate NO2 (Extended Data Fig. 6).

We constructed a split-Gal4 line to target hAB neurons. This split-Gal4 line is +; P{R72B05-p65.AD}attP40; P{VT055827-GAL4.DBD}attP2. We validated the expression of this line using Multi-Color-Flip-Out (MCFO) to visualize single-cell morphologies (Extended Data Fig. 6).

Wild animals

No wild animals were used in this study.

Field-collected samples

No field samples were collected for this study.

Ethics oversight

No ethical approval was required because experiments were performed on *Drosophila melanogaster*.

Note that full information on the approval of the study protocol must also be provided in the manuscript.